

TRIDENT-Luna

Triple-Redundant Integrated Design for Extraterrestrial Needs — lunar surface configuration

Status: design concept (not flight hardware) · **Oxygen:** 3.6 kg/day ($\approx$ 4.3 crew-equivalents) · **Power:** 2.3 kW average · **Mass:** 324 kg (excl. shared power plant)

1. What TRIDENT-Luna is

TRIDENT-Luna is the lunar configuration of the TRIDENT ISRU platform: a life-support-scale oxygen plant that prioritizes maintainability over production rate. Three parallel process strings sit on a shared thermal mass so one string can be isolated and cooled for servicing while the others keep producing (warm-swap). The lunar variant pivots the feedstock chain from the Martian atmosphere to **polar water ice**: ice-bearing regolith is mined and heated, the captured water is electrolyzed, and oxygen is delivered to life support while hydrogen is stored as a by-product.

2. What changes from Mars, and why

- **Feedstock:** no usable atmosphere on the Moon — water ice from permanently shadowed regions replaces atmospheric CO₂; steam electrolysis replaces SOEC CO₂ co-electrolysis.
- **Hydrogen:** the Mars core loop carries a hydrogen deficit supplied externally; the lunar loop produces 0.45 kg H₂/day as a by-product.
- **Methane:** carbon cannot be ISRU-closed on the Moon, so the baseline produces no methane (the mirror image of the Mars variant's open hydrogen item).
- **Power:** a 354 h equatorial night makes solar-battery storage non-viable at this load ($\sim$ 6,771 kg of Li-ion would be required; Mars needs only $\sim$ 189 kg). Baseline: an allocation from a shared 40 kWe-class fission surface power plant.
- **Dust:** no gas phase for an electrostatic precipitator — the lunar variant uses electrodynamic dust shields ($>90\%$ removal in vacuum simulant tests) plus mechanical exclusion and seals.
- **Regolith oxygen** ($\sim$ 40–45 wt% of regolith) exists everywhere but costs $\sim$ 20–25 kWh/kg O₂ to extract — carried as a contingency kit, not the baseline.

3. Interface and architecture

Stage	Function	Notes
Ice feedstock kit	mine and heat ice-bearing regolith; capture water vapor	$\sim$ 81 kg regolith/day at 5 wt% ice
Electrolysis strings $\times 3$	$2\text{ H}_2\text{O} \rightarrow 2\text{ H}_2 + \text{O}_2$ (SOEC-class, steam mode)	shared thermal mass; warm-swap
Storage	gaseous O ₂ buffer (3 days) + H ₂ by-product	liquefaction optional, not modeled
Dust management	electrodynamic shields + seals	no consumables
Power	$\sim$ 3.4 kWe allocation from shared fission plant	battery for transients only

4. Budgets (generated by tools/model.py)

Flows

Flow	Value
O ₂ delivered	3.6 kg/day
Water consumed (stoichiometric)	4.05 kg/day
Regolith processed	81.0 kg/day at 5 wt% ice
H ₂ by-product (stored)	0.45 kg/day

Energy

Stage	kWh/day
Ice mining + water capture	12.1
Electrolysis	23.6
Balance of plant	19.4
Total	55.1 kWh/day $\rightarrow$ 2.3 kW average

Mass

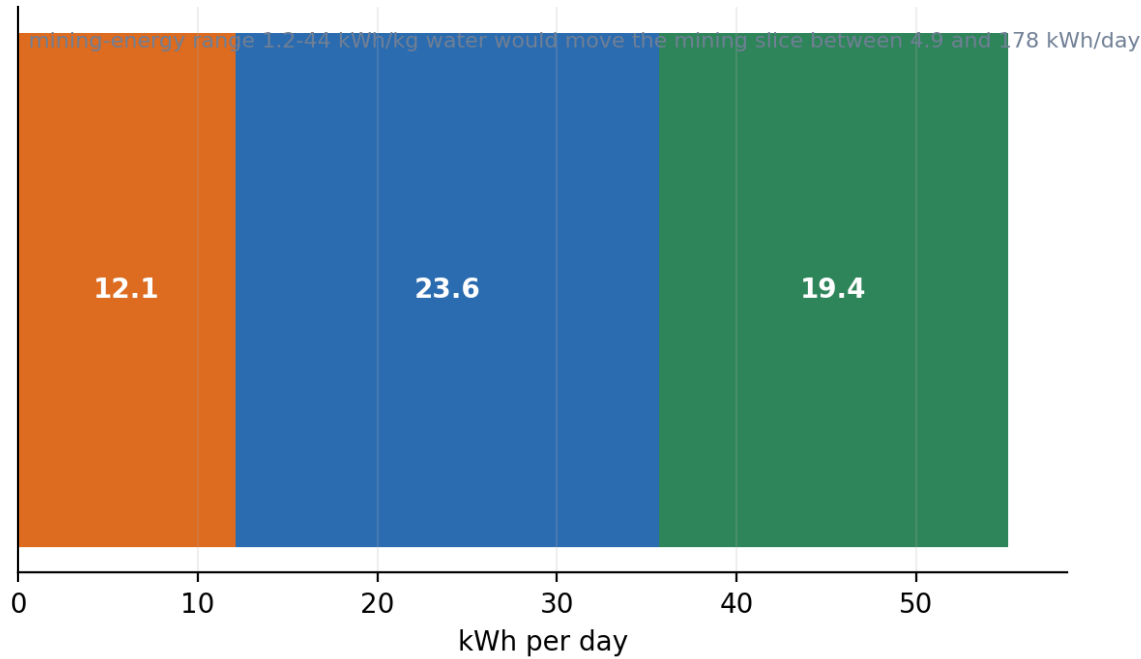
Item	kg
Processing core (strings, thermal, dust, structure, redundancy)	162

Ice feedstock kit	95
Gaseous O ₂ storage	43
Avionics + harness	24
Total	324 kg
Optional regolith-O ₂ contingency kit	+140 kg

Dominant uncertainty — ice-mining energy. Published estimates span 1.2 kWh/kg water (modeled thermal mining) to ~44 kWh/kg (lab-demonstrated, small scale). Across that band the plant's total energy moves between 47.8 and 221.1 kWh/day. The design target (55.1 kWh/day at 3 kWh/kg) assumes heat recuperation at scale and must be demonstrated.

5. Figures

TRIDENT-Luna daily energy: 55.1 kWh/day → 2.3 kW average



Concept estimates from tools/model.py (2026-10-01 run); expect $\pm 30\text{-}50\%$. Sources: variants/moon/SOURCES.md.

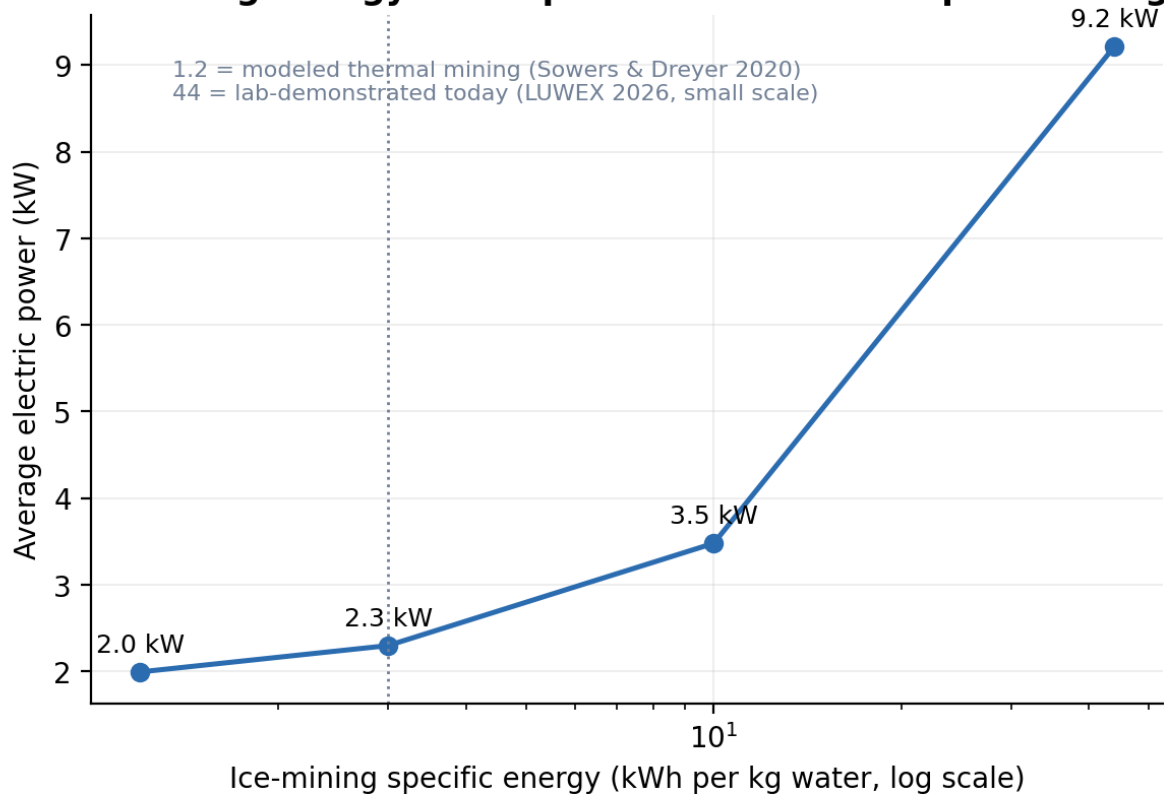
Ice mining
+ water capture

Electrolysis
(steam SOEC-class)

Balance of plant
(thermal, dust, avionics...)

Figure 1 — Daily energy by stage.

The mining-energy assumption dominates the power budget



Concept estimates from tools/model.py (2026-10-01 run); expect ± 30 -50%. Sources: variants/moon/SOURCES.md.

Figure 2 — Plant power vs ice-mining specific energy.

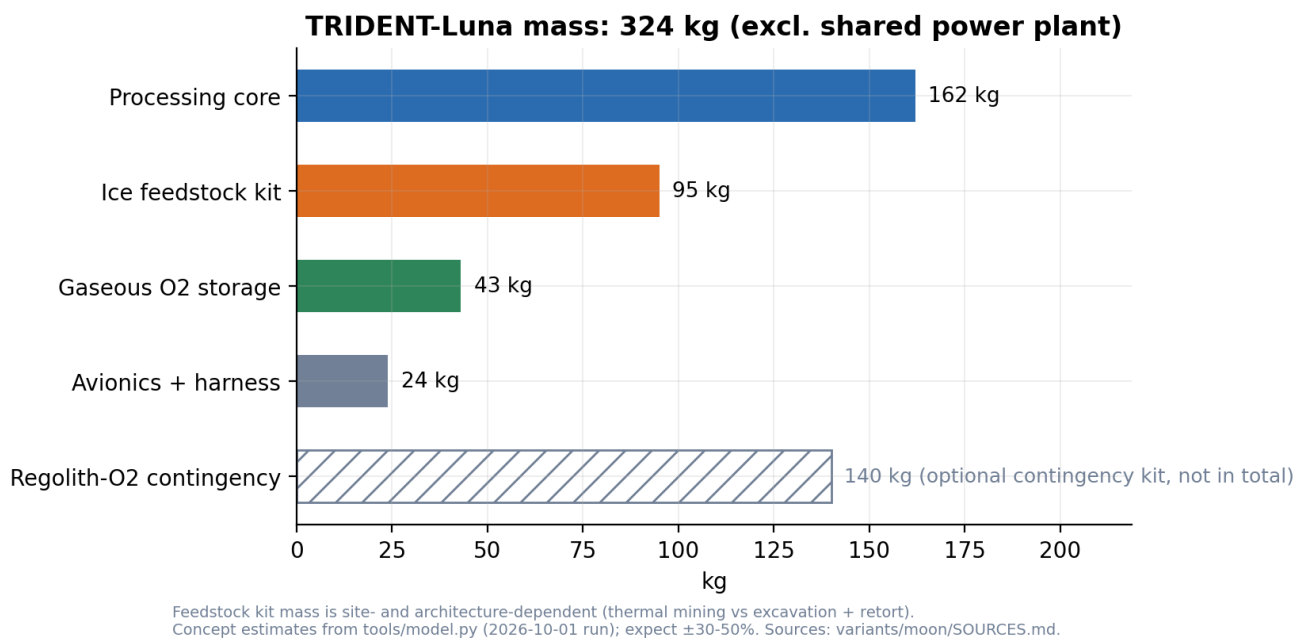


Figure 3 — Concept mass breakdown.

6. Open items

Ranked list in **ASSUMPTIONS.md**: mining energy and heat delivery (a first-order screen and a bench-test gate are now specified — see **miner-heat-transfer.md**); ice concentration; PSR-to-plant material handling; H₂ storage not modeled; fission availability; dust at the regolith interface; night thermal duty; no-methane scope check.

7. Sources

All external figures: **variants/moon/SOURCES.md** (verified 2026-10-01) — NSSDC Moon Fact Sheet; Colaprete et al. 2010 (LCROSS); Mazarico 2011 / Gläser 2014 (illumination); Paige/Hayne 2010 (PSR temperatures); NASA/ESA (regolith oxygen, EDS, MRE); DOE 2024 (electrolysis); Sowers & Dreyer 2020 and LUWEX 2026 (ice mining); NASA NTRS 20220004670 (fission surface power).

8. Disclaimer

TRIDENT-Luna is a conceptual design for research, discussion and engineering evaluation only. It is not flight hardware, has not been built or tested, and must not be treated as a construction or operational plan. All performance figures are engineering estimates; real implementation requires independent detailed design, hazard analysis, qualification testing and professional engineering oversight.

Author: Nicholas Dean Perry · design concept, October 2026 · Generated from tools/model.py (run 2026-10-01).